1 Locomotion Is a Control Problem

THE PROBLEM

Position-based locomotion is accurate under known conditions but rigid — it struggles with compliance, disturbance handling, unknown terrain, and the sim-to-real gap.

→ Torque-based control enables compliant, adaptive interaction.

POSITION CONTROL	TORQUE CONTROL
Command position	Command force
▼	▼
Rigid behavior	Compliant interaction
▼	▼
Poor adaptation	Adaptive response

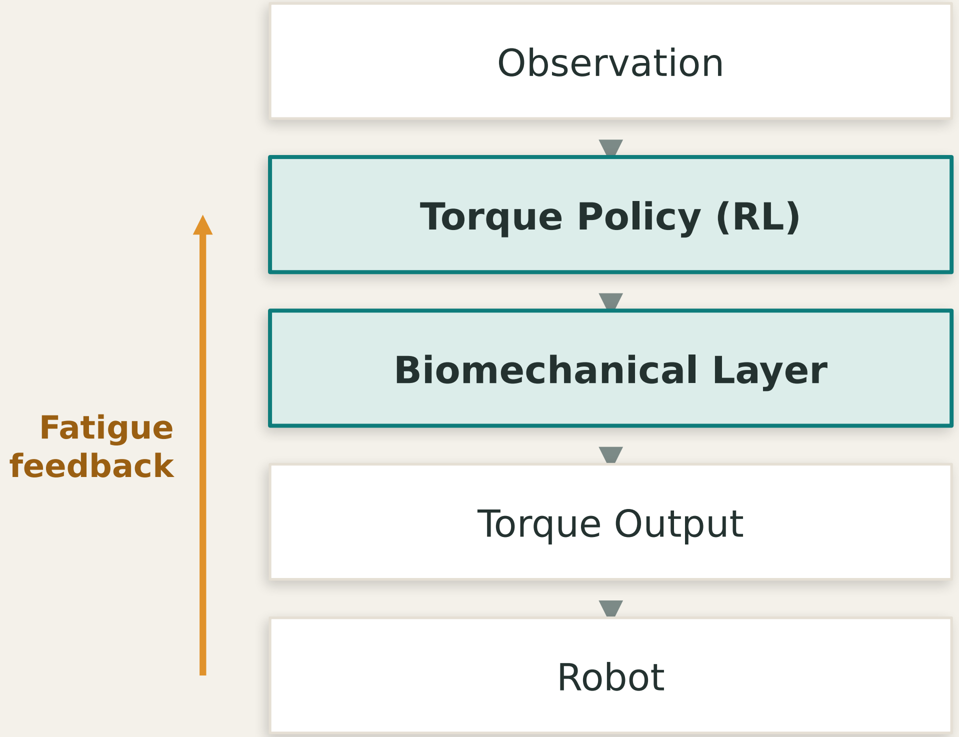
RESEARCH QUESTION

How can robots achieve safer and more adaptive locomotion in unknown environments?

Refs *SATA · Chen et al. (torque control) · Lee et al. (terrain) · Miki et al. (perceptive loco.)*

2 SATA: Bio-inspired Adaptation in Torque Control

An RL torque policy wrapped by a biomechanical layer and a growth curriculum.



M Biomechanical Model

activation · muscle · fatigue

G Growth Mechanism

torque limit · reward · frequency

Goal: smooth, safe torques that progressively unlock capability — deployed zero-shot (sim-to-real).

Refs *SATA · Hill (muscle dynamics) · Liu et al. (fatigue) · Bellegarda & Ijspeert (CPG-RL)*

3 Plan: Understand → Reproduce → Analyze

A feasible 3-phase study — we interpret SATA, we do not redesign the RL.

1 PHASE 1
Reproduce

Set up Isaac Gym on a CUDA server (container / venv) and run the official SATA training.

OUTPUT simulation running

→

2 PHASE 2
Ablation

Disable fatigue · change torque limit · modify growth schedule · vary terrain.

OUTPUT behavioral data

→

3 PHASE 3
Adaptive Interpretation

Growth → gain scheduling
Fatigue → feedback
Torque limit → adaptive constraint

OUTPUT analysis & discussion

→

4 PHASE 4 · OPTIONAL
Residual Compensation

$\tau\_total = \tau\_SATA + \tau\_comp$
Pursue only if feasible.

OUTPUT robustness in boundary cases

APPROACH

We understand and analyze adaptive behavior — we do not redesign the RL algorithm.

Refs *SATA · RL2AC · DecAP*

4 Expected Contribution

We study adaptive behavior — not propose a new RL algorithm.

EXPECTED OUTPUTS

1. Simulation reproduction
2. Behavior analysis (ablation)
3. Adaptive-control interpretation

RESEARCH QUESTIONS

RQ1
Why torque control?

RQ2
What creates adaptation?

RQ3
How does it relate to adaptive control?

RQ4
Can lightweight compensation help?

IN ONE LINE

This project investigates how bio-inspired torque control creates adaptive locomotion behaviors, and interprets these mechanisms through an adaptive control perspective.